



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Applied DevOps

Experiment 6

MTECH - School of Computer Science
CYBER SECURITY AND FORENSICS

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Objective — To explore and implement a Continuous Deployment (CD) pipeline by integrating GitHub, Jenkins, and GitHub Pages, enabling automatic build and deployment of a web application upon code changes.

Problem Statement:

Design and implement a fully automated deployment pipeline where:

- Code changes pushed to GitHub
- Trigger a pipeline in Jenkins via webhook
- The pipeline builds and deploys the application
- The application is hosted using GitHub Pages

Tools and Environment :

Development Environment

- Operating System: macOS (MacBook Air)
IDE: Visual Studio Code
- Terminal: zsh shell

Version Control

- Git: Used for tracking source code changes and managing versions
- GitHub: Cloud-based repository hosting platform
Authentication Method: SSH key-based authentication.

Automation

- Jenkins : Used to create and run CI/CD pipelines
- Ngrok : Exposes local Jenkins server to the internet and generates a public URL for webhook integration.

Task to be Performed

Task 1: Create a Sample Web Application

```
cd-demo-app - zsh - 99x33

[sheoraninfosec@Jigeshs-MacBook-Air ~ % mkdir cd-demo-app
[sheoraninfosec@Jigeshs-MacBook-Air ~ % cd cd-demo-app
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % pwd
/Users/sheoraninfosec/cd-demo-app
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % nano index.html
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ]

cd-demo-app - nano index.html - 99x33

UW PICO 5.09 File: index.html

<!DOCTYPE html>
<html>
<head>
  <title>CD Demo</title>
</head>
<body>
  <h1>Continuous Deployment Demo</h1>
</body>
</html>
```

Task 2: Initialize and Push to GitHub

```
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/sheoraninfosec/cd-demo-app/.git/
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git add .
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git commit -m "Initial commit"
[master (root-commit) 2c9a766] Initial commit
 1 file changed, 9 insertions(+)
 create mode 100644 index.html
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git branch -M master
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git remote add origin https://github.com/sheoraninfosec/cd-demo-app.git
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push -u origin master
Username for 'https://github.com': sheoraninfosec
Password for 'https://github.com': 
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/sheoraninfosec/cd-demo-app.git/'
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ls -al ~/.ssh
total 48
drwx----- 9 sheoraninfosec staff 288 25 Mar 10:21 .
drwxr-x----- 45 sheoraninfosec staff 1440 27 Mar 11:14 ..
drwx----- 3 sheoraninfosec staff 96 25 Mar 10:21 agent
-rw----- 1 sheoraninfosec staff 419 22 Jan 12:16 id_ed25519
-rw-r--r-- 1 sheoraninfosec staff 104 22 Jan 12:16 id_ed25519.pub
-rw----- 1 sheoraninfosec staff 2675 1 Nov 2023 id_rsa
-rw-r--r-- 1 sheoraninfosec staff 594 1 Nov 2023 id_rsa.pub
-rw----- 1 sheoraninfosec staff 828 1 Nov 2023 known_hosts
-rw-r--r-- 1 sheoraninfosec staff 92 1 Nov 2023 known_hosts.old
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ssh-add -l
The agent has no identities.
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ssh-add ~/.ssh/id_ed25519
Identity added: /Users/sheoraninfosec/.ssh/id_ed25519 (jsheoran9755@gmail.com)
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ssh -T git@github.com
[Hi sheoraninfosec! You've successfully authenticated, but GitHub does not provide shell access.
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push -u origin master
Username for 'https://github.com': git remote -v
Password for 'https://github.com': git remote -v
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/sheoraninfosec/cd-demo-app.git/'
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git remote set-url origin git@github.com:sheoraninfosec/cd-demo-app.git
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git remote -v
origin git@github.com:sheoraninfosec/cd-demo-app.git (fetch)
origin git@github.com:sheoraninfosec/cd-demo-app.git (push)
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 306 bytes | 153.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/cd-demo-app.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ]
```

Task 3: Set up Jenkins

```
(sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % jenkins-lts
Running from: /opt/homebrew/Cellar/jenkins-lts/2.541.3/libexec/jenkins.war
webroot: /Users/sheoraninfosec/.jenkins/war
2026-03-31 08:58:49.274+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-03-31 08:58:50.007+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-03-31 08:58:50.031+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 17.0.18+0
2026-03-31 08:58:50.230+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, d
id not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-03-31 08:58:50.250+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
2026-03-31 08:58:50.411+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /Users/sheora
ninfosec/.jenkins found at: $user.home/.jenkins
2026-03-31 08:58:51.077+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreC
ontextHandler@769a58e5{Jenkins v2.541.3,/b=file:///Users/sheoraninfosec/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$
CoreContextHandler$CoreToNestedHandler@55f8669d{STARTED}}
2026-03-31 08:58:51.081+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@65fd67
08{HTTP/1.1,(http/1.1)}{0.0.0.0:8080}
2026-03-31 08:58:51.082+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@2532b351{START
ING}
2026-03-31 08:58:51.082+0000 [id=27] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=d
isabled
2026-03-31 08:58:51.138+0000 [id=26] INFO jenkins.model.Jenkins#<init>: Starting version 2.541.3
2026-03-31 08:58:51.177+0000 [id=35] INFO jenkins.InitReactorRunner$1onAttained: Started initialization
2026-03-31 08:58:51.180+0000 [id=38] INFO jenkins.InitReactorRunner$1onAttained: Listed all plugins
2026-03-31 08:58:51.434+0000 [id=36] INFO jenkins.InitReactorRunner$1onAttained: Prepared all plugins
2026-03-31 08:58:51.435+0000 [id=45] INFO jenkins.InitReactorRunner$1onAttained: Started all plugins
2026-03-31 08:58:51.436+0000 [id=46] INFO jenkins.InitReactorRunner$1onAttained: Augmented all extensions
2026-03-31 08:58:51.498+0000 [id=47] INFO jenkins.InitReactorRunner$1onAttained: System config loaded
2026-03-31 08:58:51.498+0000 [id=34] INFO jenkins.InitReactorRunner$1onAttained: System config adapted
2026-03-31 08:58:51.498+0000 [id=33] INFO jenkins.InitReactorRunner$1onAttained: Loaded all jobs
2026-03-31 08:58:51.499+0000 [id=33] INFO jenkins.InitReactorRunner$1onAttained: Configuration for all jobs updated
2026-03-31 08:58:51.517+0000 [id=62] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-03-31 08:58:51.726+0000 [id=45] INFO jenkins.install.SetupWizard#init:
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 3e5ca5b5a16b427aa62b2fda02022bcf
[LF]>
[LF]> This may also be found at: /Users/sheoraninfosec/.jenkins/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
2026-03-31 08:58:58.140+0000 [id=38] INFO jenkins.InitReactorRunner$1onAttained: Completed initialization
2026-03-31 08:58:58.150+0000 [id=26] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-03-31 08:59:01.308+0000 [id=62] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for h
udson.tasks.Maven.MavenInstaller
2026-03-31 08:59:01.308+0000 [id=62] INFO hudson.util.Retrier#start: Performed the action check updates server succes
sfully at the attempt #1
```

Step 1: Copy the password from your terminal.

localhost:8080/login?from=%2F

Relaunch to update

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/Users/sheoraninfosec/.jenkins/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Step 2 : Install Plugins

Getting Started			
✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding
✓ Timestamper	⌚ Workspace Cleanup	⌚ Ant	⌚ Gradle
⌚ Pipeline	⌚ GitHub Branch Source	⌚ Pipeline: GitHub Groovy Libraries	⌚ Pipeline Graph View
⌚ Git	⌚ SSH Build Agents	⌚ Matrix Authorization Strategy	⌚ LDAP
⌚ Email Extension	⌚ Mailer	⌚ Dark Theme	

Step 3 : Create Admin User

Create First Admin User

Username

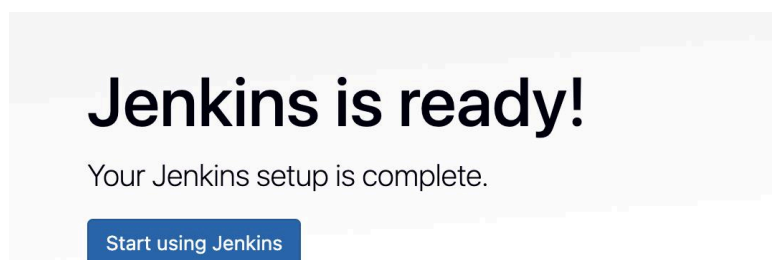
Password

Confirm password

Full name

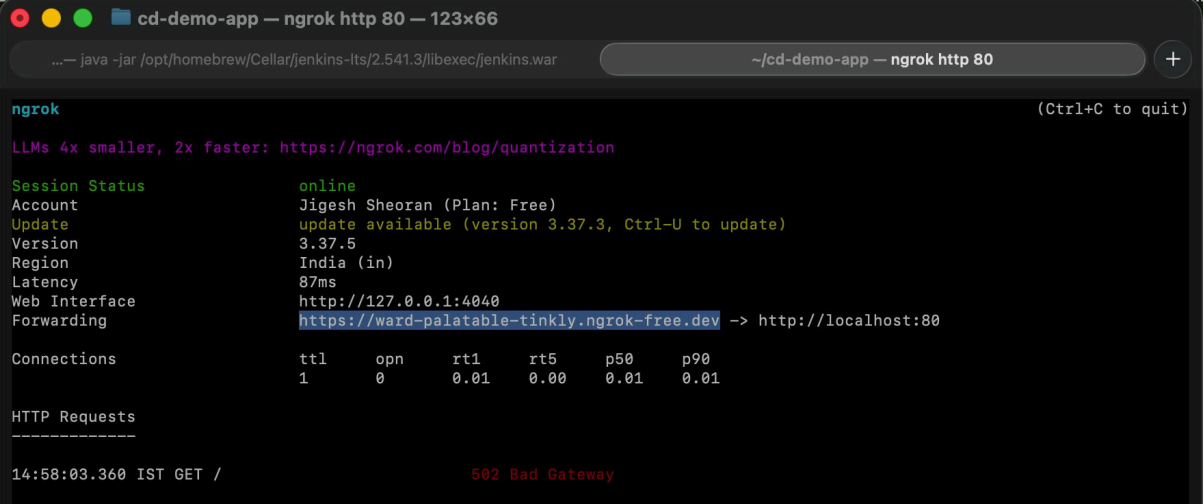
E-mail address

Step 4: Check Configuration and Finish



Task 4: Configure Webhook Integration

1. Use ngrok to expose Jenkins: **ngrok http 8080**
2. Copy the generated HTTPS URL
3. In GitHub:
 - Navigate to Settings → Webhooks
 - Add webhook: ▪ Payload URL: <ngrok-url>/github-webhook/
 - Trigger: *Push events*



The screenshot shows a terminal window titled "cd-demo-app — ngrok http 80 — 123x66". The terminal output displays the ngrok interface, including session status, account information, and a table of connections. A "502 Bad Gateway" error is visible at the bottom.

```
ngrok (Ctrl+C to quit)
LLMs 4x smaller, 2x faster: https://ngrok.com/blog/quantization

Session Status      online
Account             Jigesh Sheoran (Plan: Free)
Update              update available (version 3.37.3, Ctrl-U to update)
Version             3.37.5
Region             India (in)
Latency             87ms
Web Interface       http://127.0.0.1:4040
Forwarding           https://ward-palatable-tinkly.ngrok-free.dev -> http://localhost:80

Connections
  ttl   opn   rt1   rt5   p50   p90
    1     0   0.01  0.00  0.01  0.01

HTTP Requests
-----
14:58:03.360 IST GET /                               502 Bad Gateway
```

Generated https URL : <https://ward-palatable-tinkly.ngrok-free.dev>

Step 1: Configure Webhook in GitHub

The screenshot shows the GitHub repository settings for 'cd-demo-app' under the 'Settings' tab. The left sidebar contains a navigation menu with categories: General, Access, Code and automation, Security, and Integrations. The 'Webhooks' option is selected. The main content area is titled 'Webhooks / Add webhook'. It contains the following fields and options:

- Webhooks / Add webhook**
- General**
- Access**
- Collaborators**
- Moderation options**
- Code and automation**
- Branches**
- Tags**
- Rules**
- Actions**
- Models** (with a 'Preview' button)
- Webhooks** (selected)
- Copilot**
- Environments**
- Codespaces**
- Pages**
- Security**
- Advanced Security**
- Deploy keys**
- Secrets and variables**
- Integrations**
- GitHub Apps**
- Email notifications**

The 'Webhooks / Add webhook' section includes:

- General**
- Webhooks / Add webhook**
- General**
- Access**
- Collaborators**
- Moderation options**
- Code and automation**
- Branches**
- Tags**
- Rules**
- Actions**
- Models** (with a 'Preview' button)
- Webhooks** (selected)
- Copilot**
- Environments**
- Codespaces**
- Pages**
- Security**
- Advanced Security**
- Deploy keys**
- Secrets and variables**
- Integrations**
- GitHub Apps**
- Email notifications**

The 'Webhooks / Add webhook' section includes:

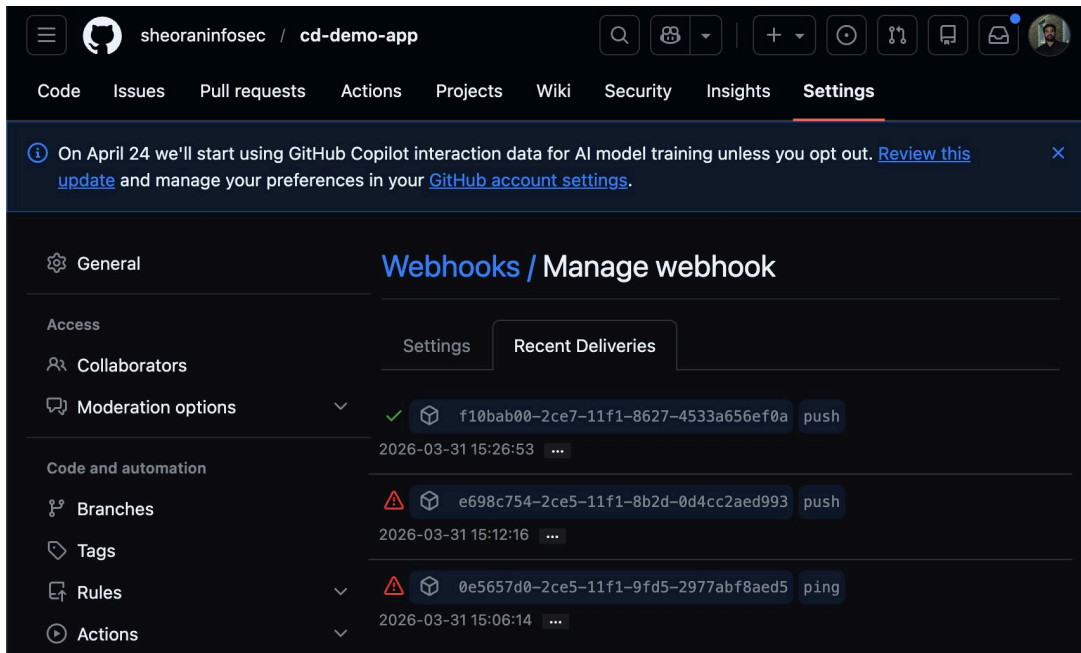
- General**
- Webhooks / Add webhook**
- General**
- Access**
- Collaborators**
- Moderation options**
- Code and automation**
- Branches**
- Tags**
- Rules**
- Actions**
- Models** (with a 'Preview' button)
- Webhooks** (selected)
- Copilot**
- Environments**
- Codespaces**
- Pages**
- Security**
- Advanced Security**
- Deploy keys**
- Secrets and variables**
- Integrations**
- GitHub Apps**
- Email notifications**

Step 2: Test the webhook.

```
Last login: Tue Mar 31 14:57:44 on ttys002
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % echo "Webhook working" >> index.html
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git add .
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git commit -m "Trigger webhook"
[master a9d0be3] Trigger webhook
1 file changed, 1 insertion(+)
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 292 bytes | 292.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:sheoraninfosec/cd-demo-app.git
2c9a766..a9d0be3 master -> master
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ]
```

This webhook failed so i reconfigured everything and then tried to test it again, hoping it would pass.

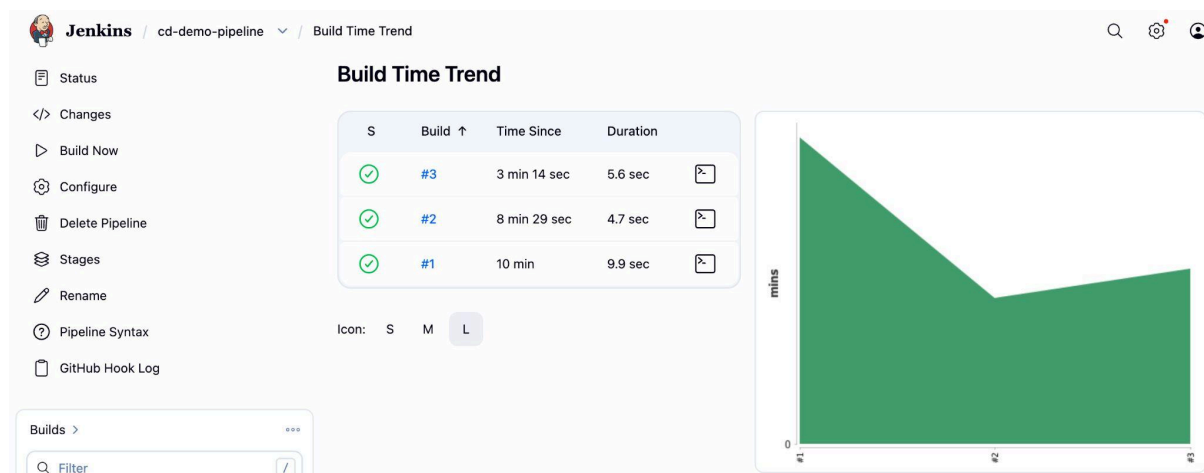
```
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git commit --allow-empty -m "trigger pipeline"
[master 4ef2a50] trigger pipeline
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push origin master
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 188 bytes | 188.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sheoraninfosec/cd-demo-app.git
a9d0be3..4ef2a50 master -> master
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % ]
```

Task 5: Create Jenkins Pipeline



Pipeline created and is executing perfectly when jenkins triggers build.



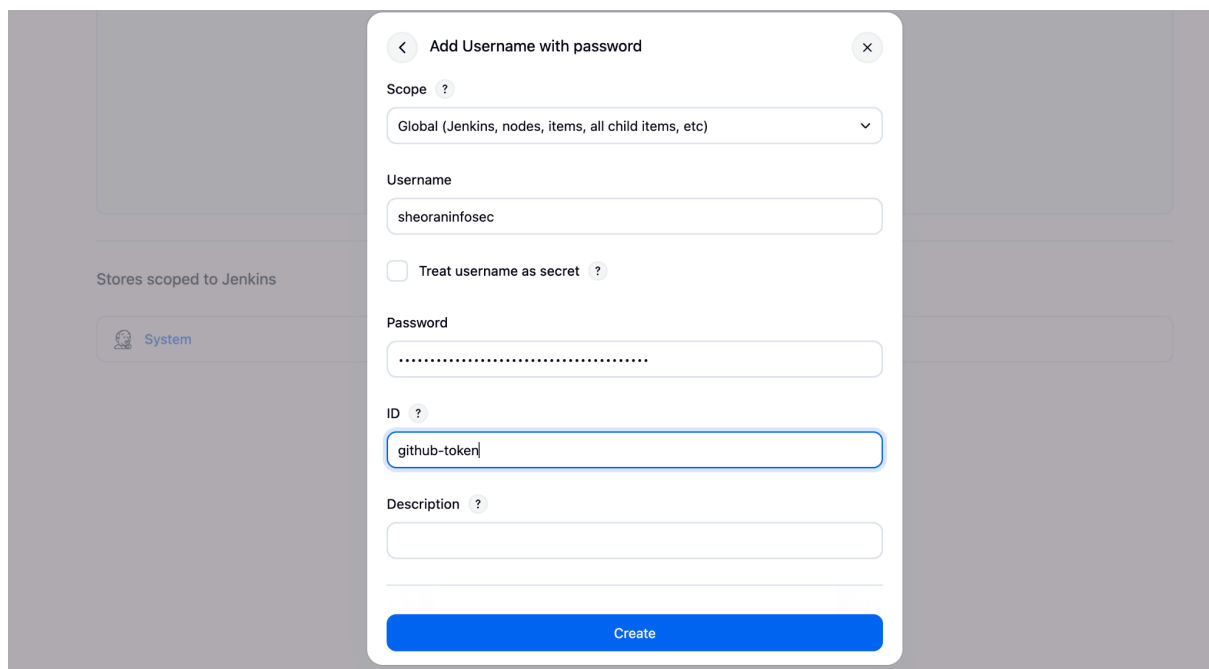
Step 1: Create **gh-pages** branch

Step 2: Enable GitHub Pages

Site is Live



Step 4: Store Token in Jenkins



The screenshot shows the 'Add Username with password' dialog in Jenkins. The 'Scope' is set to 'Global (Jenkins, nodes, items, all child items, etc)'. The 'Username' field contains 'sheoraninfosec'. The 'Treat username as secret' checkbox is unchecked. The 'Password' field is masked with dots. The 'ID' field contains 'github-token'. The 'Description' field is empty. A blue 'Create' button is at the bottom.

Stores scoped to Jenkins

System

Add Username with password

Scope ?
Global (Jenkins, nodes, items, all child items, etc)

Username
sheoraninfosec

☐ Treat username as secret ?

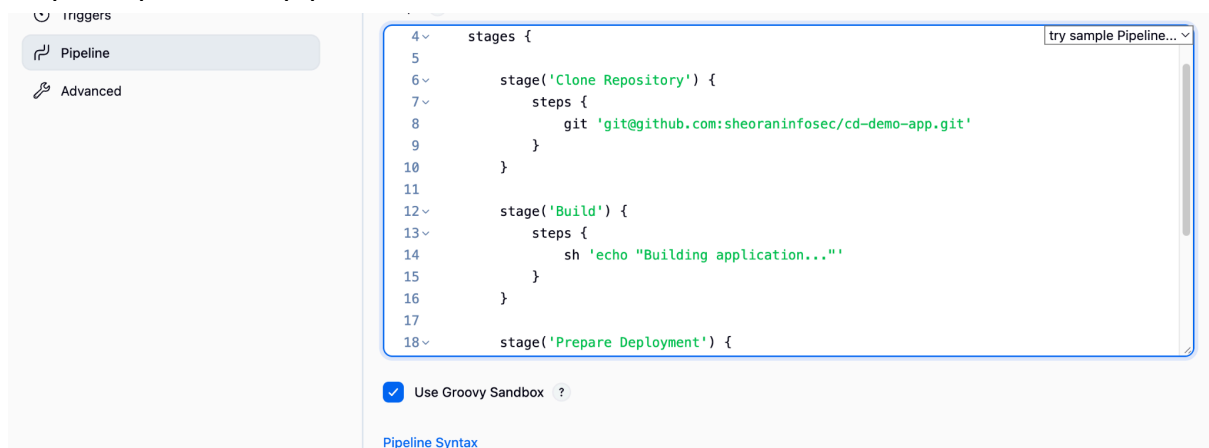
Password
.....

ID ?
github-token

Description ?

Create

Step 5: Update the pipeline



The screenshot shows the Jenkins Pipeline configuration page. The left sidebar has 'Pipeline' selected. The main area shows a pipeline script with three stages: 'Clone Repository', 'Build', and 'Prepare Deployment'. The 'Clone Repository' stage uses the 'git' step with the repository URL 'git@github.com:sheoraninfosec/cd-demo-app.git'. The 'Build' stage uses the 'sh' step with the command 'echo "Building application..."'. The 'Prepare Deployment' stage is currently empty. The 'Use Groovy Sandbox' checkbox is checked. A 'Pipeline Syntax' link is at the bottom.

triggers

Pipeline

Advanced

```
4 stages {  
5  
6 stage('Clone Repository') {  
7 steps {  
8 git 'git@github.com:sheoraninfosec/cd-demo-app.git'  
9 }  
10 }  
11  
12 stage('Build') {  
13 steps {  
14 sh 'echo "Building application..."'  
15 }  
16 }  
17  
18 stage('Prepare Deployment') {
```

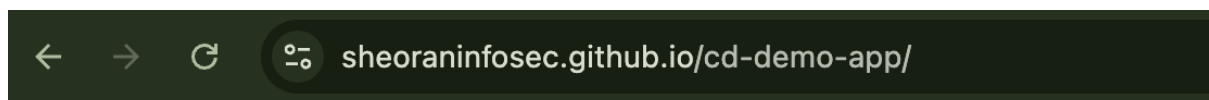
☒ Use Groovy Sandbox ?

[Pipeline Syntax](#)

Step 6: Trigger deployment

```
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git commit --allow-empty -m "deploy test"  
[gh-pages 0944141] deploy test  
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push origin master  
Everything up-to-date  
sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app %
```

Step 7: Check live website



The screenshot shows a web browser address bar with the URL 'sheoraninfosec.github.io/cd-demo-app/'.

← → ↻ 🔍 sheoraninfosec.github.io/cd-demo-app/

initial

Task 7: Enable GitHub Pages (already done in task 6)

Your site is live at <https://sheoraninfosec.github.io/cd-demo-app/>
Last [deployed](#) by  [sheoraninfosec](#) 3 minutes ago

[Visit site](#) [Unpublish site](#)

Build and deployment

Source

Deploy from a branch ▾

Branch

Your GitHub Pages site is currently being built from the gh-pages branch. [Learn more about configuring the publishing source for your site.](#)

 gh-pages ▾

 / (root) ▾

Save

Task 8 : Test Continuous Deployment

Step 1 : Modify the index.html file

```
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % nano index.html  
sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % █
```

Step 2: Commit and push changes

```
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git add .  
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git commit -m "Updated website content"  
[gh-pages 3ebf726] Updated website content  
1 file changed, 1 insertion(+), 1 deletion(-)  
[sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % git push origin master  
Everything up-to-date  
sheoraninfosec@Jigeshs-MacBook-Air cd-demo-app % █
```

Observation

- A sample web application was successfully created and initialized using Git
- The project repository was pushed to GitHub and managed remotely
- Jenkins was installed and configured to create a pipeline
- Webhook integration using ngrok enabled external communication with local Jenkins
- On every code push, the webhook triggered the Jenkins pipeline automatically
- The pipeline successfully performed cloning, build, and deployment steps
- Deployment was carried out by pushing files to the **gh-pages** branch
- GitHub Pages hosted the application and made it accessible via a public URL
- Changes made in the code were automatically reflected on the live website without manual deployment

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